

Concept Learning (Simplified)

What is Concept Learning?

It's like learning **rules** from examples so that you can **predict or classify** new things.

Real-Life Example:

You see different weather conditions (sunny, rainy, humid) and whether your friend enjoys playing sports in those. You try to **learn the pattern** (rule) behind when they say "Yes" to playing.

Rule-Based Learning

 We teach the computer using **rules** like:

IF it's sunny AND warm THEN play sports.

This is called an **If-Then rule**.

Find-S Algorithm (Easy Version)

 **Goal:** Learn from only the **positive examples** ("Yes" cases).

Steps:

1. Start with a blank or very specific guess.
2. For every "Yes" example:
 - Make the rule **a little more general** (just enough to include that example).
3. Ignore "No" examples.

 Example:

- First example: Sunny, Warm, Normal, Strong, Warm, Same → Yes
- Rule starts as: Sunny, Warm, Normal, Strong, Warm, Same
- Next example: Sunny, Warm, High, Strong, Warm, Same → Yes
- Rule changes: Sunny, Warm, ?, Strong, Warm, Same

(Only "Humidity" changed, so it's replaced with ? meaning "any value is OK.")



Candidate-Elimination Algorithm (Easy Version)



Goal: Use **both** "Yes" and "No" examples to find the best rule.

We keep two boundaries:

- S = Most specific rules
- G = Most general rules

With each example:

- If it's "Yes", we **make S more general**
- If it's "No", we **make G more specific**

Eventually, both will **close in** on the right rule.



Imagine a funnel narrowing down to the correct answer.



Decision Tree (Like 20 Questions Game!)

It's like asking questions step-by-step to reach a decision:

mathematica

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```

└─ Humidity = Normal?
    └─ Yes → Play
    └─ No → Don't Play
  
```

Each path from top to bottom becomes a **rule**.



Python Code (Find-S)

Here's a small program that learns rules from "Yes" examples:

python

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def find_s(examples): h = examples[0][:-1] # First example (remove
Yes/No) for example in examples: if example[-1] == "Yes": for i in
range(len(h)): if example[i] != h[i]: h[i] = "?" # Generalize
return h
  
```



Easy Summary

Concept

Concept Learning

Find-S

Candidate-Elimination

Rule

Decision Tree





If This Comes in Exam...

- ◆ **Define the algorithm** (use simple words).
- ◆ **Give 1-2 short examples.**
- ◆ **Draw a simple diagram** (for bonus marks!).
- ◆ **Keep points short & easy to read.**
- ◆ **Mention one advantage** (like: "easy to understand").